GraphTheorySymbol Memo
v1.7.0 2026/09/21

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
`\GTSadjacency/`	`\GTSa/`		u v
`\GTSadjacencyRelation/`	`\GTSaR/`		u v
`\GTSunadjacency/`	`\GTSu/`		u v
`\GTSunadjacencyRelation/`	`\GTSuR/`		u v
`\GTSdirectedAdjacencyUp/`	`\GTSdAU/`		u v
`\GTSdirectedAdjacencyUpRelation/`	`\GTSdAUR/`		u v
`\GTSnegatedDirectedAdjacencyUp/`	`\GTSnDAU/`		u v
`\GTSnegatedDirectedAdjacencyUpRelation/`	`\GTSnDAUR/`		u v
`\GTSdirectedAdjacencyDown/`	`\GTSdAD/`		u v
`\GTSdirectedAdjacencyDownRelation/`	`\GTSdADR/`		u v
`\GTSnegatedDirectedAdjacencyDown/`	`\GTSnDAD/`		u v
`\GTSnegatedDirectedAdjacencyDownRelation/`	`\GTSnDADR/`		u v
`\GTSincidence/`	`\GTSi/`		v e
`\GTSincidenceRelation/`	`\GTSiR/`		v e
`\GTSnegatedIncidence/`	`\GTSnI/`		v e
`\GTSnegatedIncidenceRelation/`	`\GTSnIR/`		v e
`\GTSoutIncidence/`	`\GTSoI/`		v e
`\GTSoutIncidenceRelation/`	`\GTSoIR/`		v e
`\GTSnegatedOutIncidence/`	`\GTSnOI/`		v e
`\GTSnegatedOutIncidenceRelation/`	`\GTSnOIR/`		v e
`\GTSinIncidence/`	`\GTSiI/`		v e
`\GTSinIncidenceRelation/`	`\GTSiIR/`		v e
`\GTSnegatedInIncidence/`	`\GTSnII/`		v e
`\GTSnegatedInIncidenceRelation/`	`\GTSnIIR/`		v e

Long name	Short name	Result	In context
\GTSadjacencyMonochrome/	\GTSaM/		u v
\GTSadjacencyMonochromeRelation/	\GTSaMR/		u v
\GTSunadjacencyMonochrome/	\GTSuM/		u v
\GTSunadjacencyMonochromeRelation/	\GTSuMR/		u v
\GTSdirectedAdjacencyUpMonochrome/	\GTSdAUM/		u v
\GTSdirectedAdjacencyUpMonochromeRelation/	\GTSdAUMR/		u v
\GTSnegatedDirectedAdjacencyUpMonochrome/	\GTSnDAUM/		u v
\GTSnegatedDirectedAdjacencyUpMonochromeRelation/	\GTSnDAUMR/		u v
\GTSdirectedAdjacencyDownMonochrome/	\GTSdADM/		u v
\GTSdirectedAdjacencyDownMonochromeRelation/	\GTSdADMR/		u v
\GTSnegatedDirectedAdjacencyDownMonochrome/	\GTSnDADM/		u v
\GTSnegatedDirectedAdjacencyDownMonochromeRelation/	\GTSnDADMR/		u v
\GTSincidenceMonochrome/	\GTSiM/		v e
\GTSincidenceMonochromeRelation/	\GTSiMR/		v e
\GTSnegatedIncidenceMonochrome/	\GTSnIM/		v e
\GTSnegatedIncidenceMonochromeRelation/	\GTSnIMR/		v e
\GTSoutIncidenceMonochrome/	\GTSoIM/		v e
\GTSoutIncidenceMonochromeRelation/	\GTSoIMR/		v e
\GTSnegatedOutIncidenceMonochrome/	\GTSnOIM/		v e
\GTSnegatedOutIncidenceMonochromeRelation/	\GTSnOIMR/		v e
\GTSinIncidenceMonochrome/	\GTSiIM/		v e
\GTSinIncidenceMonochromeRelation/	\GTSiIMR/		v e
\GTSnegatedInIncidenceMonochrome/	\GTSnIIM/		v e
\GTSnegatedInIncidenceMonochromeRelation/	\GTSnIIMR/		v e

\GTSsetMainColor{blue}

Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u v
\GTSadjacencyRelation/	\GTSaR/		u v
\GTSunadjacency/	\GTSu/		u v
\GTSunadjacencyRelation/	\GTSuR/		u v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u v
\GTSincidence/	\GTSi/		v e
\GTSincidenceRelation/	\GTSiR/		v e
\GTSnegatedIncidence/	\GTSnI/		v e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v e
\GTSoutIncidence/	\GTSoI/		v e
\GTSoutIncidenceRelation/	\GTSoIR/		v e
\GTSnegatedOutIncidence/	\GTSnOI/		v e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v e
\GTSinIncidence/	\GTSiI/		v e
\GTSinIncidenceRelation/	\GTSiIR/		v e
\GTSnegatedInIncidence/	\GTSnII/		v e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v e

Long name	Short name	Result	In context
`\GTSmetaNot/`	`\GTSmN/`	$\overset{\mathrm{m}}{\neg}$	$\overset{\mathrm{m}}{\neg}\circ\!\!\downarrow\!\!\circ$
`\GTSmetaAnd/`	`\GTSmA/`	$\overset{\mathrm{m}}{\wedge}$	$\circ\!\!\downarrow\!\!\overset{\mathrm{m}}{\wedge}\!\!\downarrow\!\!\circ$
`\GTSmetaOr/`	`\GTSmO/`	$\overset{\mathrm{m}}{\vee}$	$\circ\!\!\downarrow\!\!\overset{\mathrm{m}}{\vee}\!\!\downarrow\!\!\circ$

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:
$\forall u, v, u\ [\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]\ v$.

Tournament adjacency with maybe some colored edge also, who knows?:
$\forall u, v, u\ \circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\ v$.

Directed graph:
$\forall u, v, u\ [\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]\ v$.

Oriented graph:
$\forall u, v, u\ [\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]\ v$.

Tournament adjacency:
$[\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]$ is obtained with `\mathrel{[\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/]}`.

Tournament adjacency with maybe some colored edge also, who knows?:
$\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ$ is obtained with `\mathrel{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/}`.

Directed graph:
$[\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]$ is obtained with
`\mathrel{[\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/`
`\GTSmO/\GTSdAD/\GTSdAU/\GTSmO/\GTSnDAD/\GTSnDAU/]}`.

Oriented graph:
$[\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]$ is obtained with
`\mathrel{[\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/\GTSmO/\GTSnDAD/\GTSnDAU/]}`.

Prefix notation:
$\exists u, v, u\ [\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ]\ v \Leftrightarrow \exists u, v, u\ [\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]\ v$.

$\exists v, e, v\ [\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ]\ e \Leftrightarrow \exists v, e, v\ [\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ]\ e$.

$[\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ]$ is obtained with
`\mathrel{[\GTSmO/\GTSxa[mC=red,d]\GTSxa[mC=green,d]\GTSxa[mC=blue,u]]}`.

$[\overset{\mathrm{m}}{\vee}\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ\!\!\downarrow\!\!\circ]$ is obtained with
`\mathrel{[\GTSmO/\GTSxi[mC=red,i]\GTSxi[mC=green,i]\GTSxi[mC=blue,o]]}`.

Apart from these symbol macros, versatile macros to call them all are available with options:
`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,
`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,
`\GTSxml[]`, or `\GTSxmetalogic[]` for meta-logic symbols,
`\GTSx[]{}` for all symbols
(`\GTSx[]{a} = \GTSxa[]`, `\GTSx[]{i} = \GTSxi[]`, `\GTSx[]{m} = \GTSxml[]`).
See Detailed usage section for full syntax of these macros.